A. Complete Letter-Level Results

Table 3 includes every letter, not only selected gains or regressions. Metrics are percentages; Δn is the change in correct true-letter images. "Base" is the project-profile SVM with 820 Shin inputs; "Angles" adds the 15 continuous measurements. Support varies because detection filtering is not uniform across letters. Precision measures the correctness of predictions of a letter, while recall measures recovery of its true images.

Table 3. All-letter pooled OOF comparison.

		Precision (%)		Recall (%)		F1 (%)		
Letter	Images	Base	Angles	Base	Angles	Base	Angles	Δn
A	2,188	98.32	98.00	98.67	98.72	98.49	98.36	+1
B	2,204	99.28	99.14	99.77	99.73	99.52	99.43	-1
C	1,912	99.17	99.07	99.84	99.84	99.50	99.45	+0
D	2,456	99.01	98.81	97.68	97.68	98.34	98.24	+0
E	2,308	98.24	98.58	99.00	99.22	98.62	98.90	+5
F	2,859	99.23	99.26	98.60	98.57	98.91	98.91	-1
G	2,430	98.89	98.94	99.42	99.42	99.16	99.18	+0
H	2,371	98.81	98.64	98.14	98.19	98.48	98.41	+1
I	2,380	97.29	97.46	96.68	96.76	96.99	97.11	+2
J	2,576	98.07	98.19	96.78	96.78	97.42	97.48	+0
K	2,701	98.92	99.18	98.26	98.37	98.59	98.77	+3
L	2,516	99.72	99.80	99.64	99.60	99.68	99.70	-1
M	1,602	88.74	89.11	90.01	91.95	89.37	90.51	+31
N	1,280	88.89	91.32	88.12	87.97	88.51	89.61	-2
O	2,264	96.86	96.73	98.19	97.88	97.52	97.30	-7
P	2,041	97.96	97.82	98.63	98.78	98.29	98.29	+3
Q	2,106	98.75	98.94	97.72	97.72	98.23	98.33	+0
R	2,543	96.23	96.87	93.43	93.83	94.81	95.33	+10
S	2,559	96.87	97.15	98.09	98.40	97.48	97.77	+8
T	2,354	99.14	99.06	98.43	98.47	98.78	98.76	+1
U	2,513	92.05	92.11	96.70	96.58	94.31	94.29	-3
V	2,552	98.20	98.32	98.24	98.43	98.22	98.37	+5
W	2,456	98.41	98.21	98.25	98.33	98.33	98.27	+2
X	2,157	96.50	96.86	97.03	97.22	96.76	97.04	+4
Y	2,580	99.46	99.50	99.46	99.38	99.46	99.44	-2
Z	2,343	99.09	98.96	97.44	97.78	98.26	98.37	+8

The remaining angle-model errors include R→U (142), N→M (141), M→N (102), U→R (44), D→O (41), Q→P (41) and O→D (19). These counts suggest several candidate investigations; prioritizing only D/O would ignore larger residual error sources. Full ordered error pairs are retained in `results/angles/confusions/all_error_pairs.csv`.

B. Full Confusion Matrices

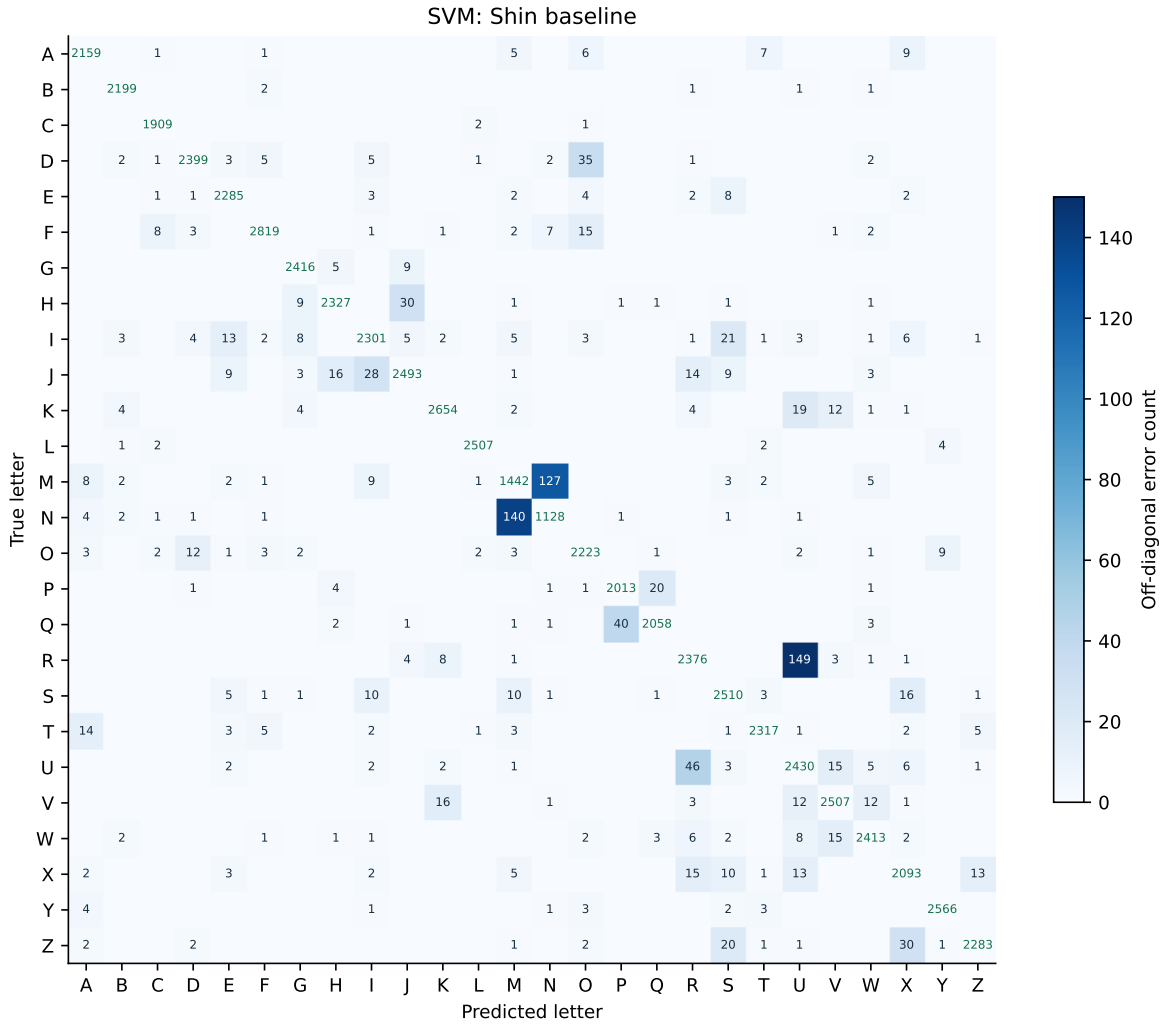


Figure 3. SVM baseline OOF confusion counts for all 26 letters. Diagonal counts are green; shading is restricted to off-diagonal errors to keep smaller confusions visible. Rows are true labels and columns predictions.

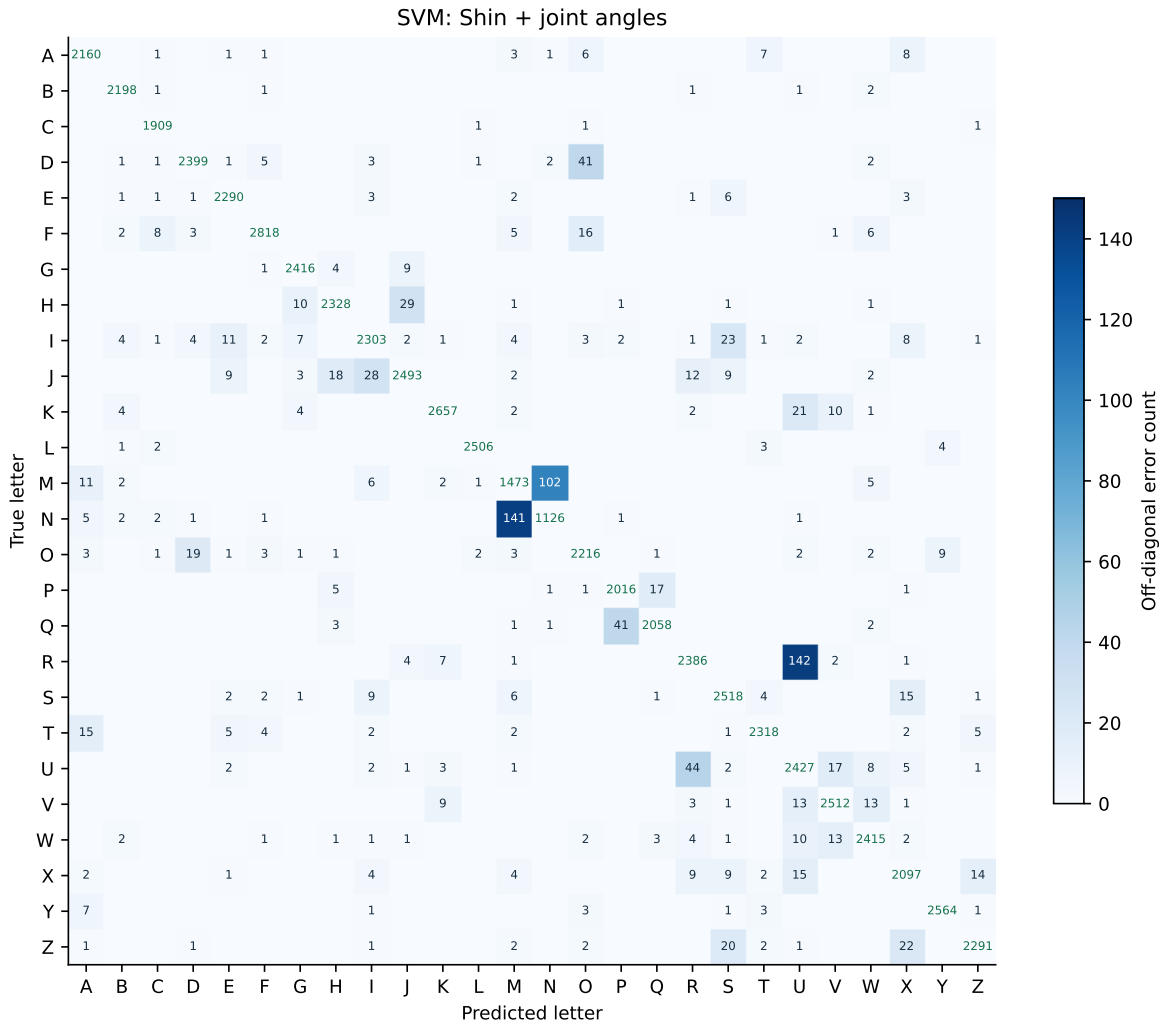


Figure 4. Same images and folds with continuous angles added. Compare, for example, the M/N, R/U and D/O cells with the baseline matrix. Counts, rather than rounded percentages alone, expose small regressions.

C. Qualitative Examples and Feature Definitions

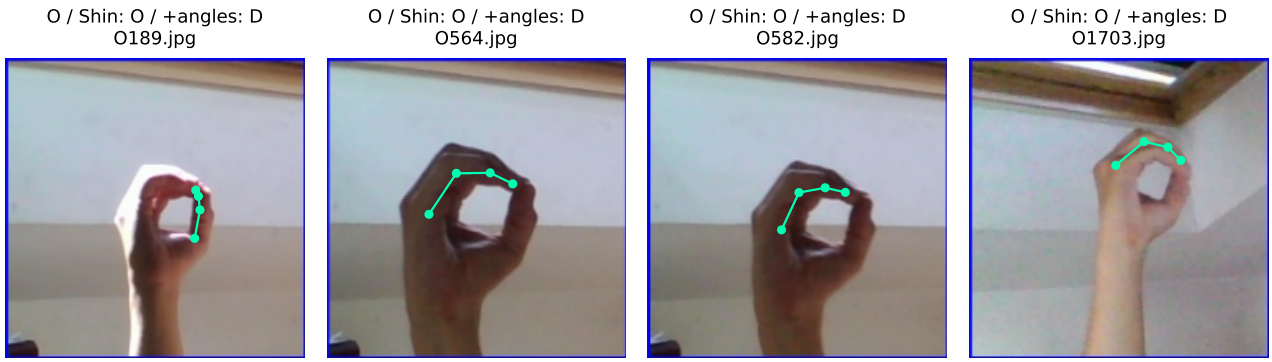


Figure 5. Four actual Kaggle O images newly classified as D after adding angles; the index-finger landmarks are overlaid. These are static dataset errors, not recorded webcam rotation tests. Dataset labels are retained; no claim of manually revalidated signing correctness is made.

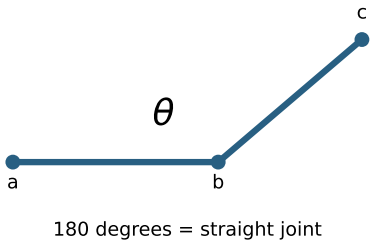


Figure 6. Interior-angle convention for a joint centred at b. A straight joint approaches 180°; flexion reduces the angle.

Table 4. Landmark triplets for the 14 interior angles, using MediaPipe indices.

Finger	MCP	PIP	DIP / IP
Thumb	(1,2,3)	—	IP: (2,3,4)
Index	(0,5,6)	(5,6,7)	(6,7,8)
Middle	(0,9,10)	(9,10,11)	(10,11,12)
Ring	(0,13,14)	(13,14,15)	(14,15,16)
Little	(0,17,18)	(17,18,19)	(18,19,20)

The fifteenth input is signed thumb-MCP deviation from the palm-oriented geometry implementation. The complete measured feature schema, unknown-state handling and thresholds are stored in `research/schema.py`, `research/geometry.py` and `config/thresholds.json`. The workbook parser preserves slash-separated alternatives as allowed sets. The user's workbook is unchanged; `config/proposal_changes.json` contains the 49 proposed relaxations. They are applied in memory for the historical compatibility-score configuration, without retaining a second spreadsheet.

For U/V/K-style spread, fingers must be sufficiently separated and approximately coplanar; they need not be strictly parallel. Crossing detection uses a palm-oriented swap/depth proxy rather than assuming any two projected image lines prove a crossing. These engineering rules are auditable hypotheses and are not validated ground-truth annotations of anatomy.

D. Paired Statistics and Weighting Details

Relative to the baseline, adding angles fixes 143 formerly incorrect images and loses 76 formerly correct images, a net gain of 67. The descriptive accuracy change is 0.1112 percentage points. Six filename blocks give the changes below. Resampling these blocks with replacement 10,000 times (seed 42) gives a central 95% sensitivity range of 0.0374–0.1947 percentage points. This is a descriptive stability check, not a valid independent-test confidence interval: there are only six non-person groups, possible cross-block correlation, overlapping training sets and exploratory model selection. We do not report a confirmatory p-value. The primary weighting study evaluated 0.25/0.5/1/2 multipliers for continuous angles,

Table 5. Paired angle-model changes by filename block.

Block	Images	Net correct	Change (pp)
0000	10,504	+4	+0.0381
0001	11,399	+27	+0.2369
0002	12,101	+2	+0.0165
0003	12,255	+5	+0.0408
0004	10,009	+23	+0.2298
0005	3,983	+6	+0.1506

categorical encodings and compatibility scores, with a single coordinate pass. In the targeted study, nested PIP weights were 2, 2, 0.5, 0.5 and 0.5 over the five outer folds. The selected MCP/PIP/DIP triples were (0.5,2,1), (1,2,0.5), (0.5,0.5,1), (2,0.5,1) and (1,0.5,1). These varying choices further caution against a universal rule that the index finger should receive higher weight.

Table 6. Targeted SVM weighting on the 835-input angle configuration. Errors are directed true-label to prediction counts.

Configuration	Accuracy (%)	Macro-F1	O→D	N→M
Unweighted	97.74776	.975244	19	141
PIP weight 2	97.74941	.975241	23	143
Nested PIP	97.75273	.975265	16	143
Nested MCP/PIP/DIP	97.73946	.975073	15	146

E. Verification and Reproducibility

The source repository retains one training dataset, the user's workbook, fixed thresholds and proposal changes, the compact standalone implementation, necessary study/live tools and tests, and compact measured evidence. Bulky model checkpoints, superseded datasets and obsolete interfaces are moved to a separate reversible archive; none is needed to interpret the reported tables. A central cached input matrix and sample/group metadata live inside the retained dataset directory. `REPRODUCE.md` documents setup, inventory validation, rebuilding descriptors from frozen landmarks, model comparisons, permutation analysis, ablations, tuning and report compilation.

The verification checks reconcile stored predictions with pooled accuracy, macro-F1, per-letter metrics and confusion totals; check one OOF prediction per image per model/variant; verify shared fold assignments and disjoint outer groups; and test feature widths, angle conventions, slash parsing, train-only preprocessing and index-weight application. Reconstructed descriptors are compared with the historical numeric cache. These checks support computational consistency; they do not validate source labels, estimated 3D anatomy or generalization to unseen signers. Large research experiments were completed before cleanup; refactoring preserves their numerical inputs and fixed profiles rather than silently reporting newly selected runs.

F. Rubric Coverage and Outstanding Research

Rubric component	Coverage and limits
Abstract	Motivation, objective, methods, numerical outcomes and scope.
Introduction	Field, motivation, contribution, questions and structure.
Related work	Baseline geometry, landmark estimation, alternative sensing/classification, permutation reliance and selection bias.
Methods	Representations, angle equation, model objectives, weighting and preprocessing.
Experiments/analysis	Matched baseline, dataset filtering, ablations, two algorithms, nested weighting, per-letter errors and descriptive paired analysis. No untouched final test or confirmatory significance claim.
Discussion	Meaning, strengths, limitations, ethics and AI-generation disclosure.
Conclusions	Modest observed angle benefit and prospective person/viewpoint-separated evaluation.

The main report now answers the explicit-feature comparison question through RQ1–RQ3. Later webcam experiments are supporting diagnostics, not unfinished conditions for that answer. A prospective signer-separated evaluation and independent review of the workbook remain valuable follow-ups. The detailed rubric assessment is retained separately in `paper/RUBRIC_REVIEW.md`.

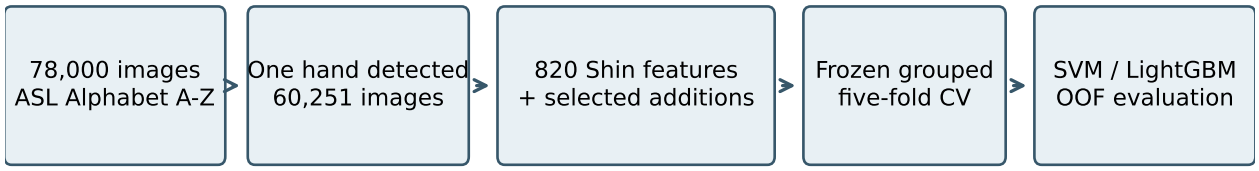


Figure 7. Evaluation pipeline. Every compared model uses the same 60,251 accepted images and frozen outer folds. Detection coverage and conditional classification accuracy are separate quantities.

G. Technical Method Details

G.1. Landmarks and Baseline

Images are processed with MediaPipe Hand Landmarker. The intended extraction settings are IMAGE mode, up to two hands, and detection/presence thresholds of 0.5; only exactly one accepted hand is used. Historical records contain the MediaPipe version and landmarks, but not the threshold values. Thus default-setting provenance cannot be independently confirmed for the historical cache. We preserve these landmarks for all comparisons rather than rerun detection differently per model.

The Shin implementation follows the paper's pair list: 190 pairwise distance features and 630 axis-direction features, for 820 features overall. Distance features are normalized per image; direction features are cosine-based encodings of inter-landmark vectors relative to axes. The local implementation preserves the paper's listed pair exclusions. Features use image landmarks for the baseline and estimated world landmarks for the added hand geometry.

G.2. Interpretable Additions

For consecutive landmarks a, b, c, an interior bending angle is

$$\theta(a,b,c) = \arccos \frac{(a-b)^\top (c-b)}{\|a-b\|\,\|c-b\|} \frac{180}{\pi}. \tag{1}$$

A straight joint approaches 180°. We use thumb MCP/IP and MCP/PIP/DIP for the other four fingers: 14 interior angles. A signed thumb-MCP deviation adds a fifteenth measurement. MCP measurements are landmark-based proxies, not calibrated clinical joint measurements. The retained phrase "angles" below always means all 15 additions, not a fourteenth-angle-only experiment.

Categorical features encode bent, partly bent and extended joints; a binary B-shaped/non-B-shaped thumb MCP; and spread, suspected crossing and thumb contact relations. An unknown state is retained. Thresholds are fixed across folds. Non-thumb flexion uses $\leq 130^\circ$ for bent and $\geq 150^\circ$ for extended, with intermediate values partly bent. Finger spread requires separation in approximately the same plane; splaying is allowed. Crossing and contact are geometric proxies and remain sensitive to occlusion and depth estimates.

The workbook gives the allowed categories for each letter. A slash means a set of acceptable alternatives, not a new combined class. One-hot measured categories yield 70 inputs. For each candidate letter ℓ, a compatibility score averages the proportion of specified workbook cells whose allowed set contains the measured category. This produces 26 scores. All 26 are computed for every image without accessing its true label. A fixed 49-cell slash-relaxation proposal was developed in earlier exploratory work; its changes are preserved separately from the user's original workbook. This reference design was not nested within the later evaluation.

G.3. Models, Scaling and Weighting

The primary SVM uses an RBF kernel, $C = 10$, `gamma=scale` and balanced class weights. This differs from the paper-profile $C = 1000$ with no class balancing. Its soft-margin objective balances margin size against weighted classification slack. For input blocks z_b and weights w_b, the RBF similarity is

$$K(x, x') = \exp\left(-\gamma \sum_b w_b^2 \|z_b(x) - z_b(x')\|^2\right). \tag{2}$$

For each binary letter pair, the kernel SVM minimizes the regularized hinge loss

$$\frac{1}{2}\|w\|^2 + C \sum_i \omega_i \max(0, 1 - y_i(w^\top \phi(x_i) + b)),$$

where $y_i \in \{-1, 1\}$, ϕ is the kernel feature mapping and ω_i combines class and sample weights. The multiclass decision uses pairwise voting. LightGBM instead optimizes multiclass log loss with tree-complexity control. These objectives were not modified to prefer individual letters.

Thus a multiplier of two contributes a factor of four inside the squared distance, while `gamma=scale` also changes with the input variance. Angle median imputation and standardization are fitted within each training fold; baseline descriptors and categorical/compatibility values pass through unchanged. No missing angles occurred in the retained cache.

LightGBM optimizes multiclass log loss through boosted trees, using 100 estimators, 31 leaves, learning rate 0.1 and minimum child size 20. Its fixed project profile is not a replication of the paper's tuned LightGBM settings. Group weighting searches use inner-fold macro-F1, breaking ties by accuracy. They make one coordinate pass, not an exhaustive global search.

H. Earlier Experimental Settings and Results

H.1. Implementation and Experimental Sequence

Experiments use Python 3.12, MediaPipe 0.10.35, scikit-learn 1.8.0 and LightGBM 4.7.0. An earlier three-fold pilot tested 81 continuous/binary finger-geometry descriptors, alone and combined with baseline descriptors. This differs from the later 70-column categorical encoding and is not mixed into the five-fold table. The pilot feature-only model reached 93.71% mean fold accuracy, versus 97.43% for the baseline; combining them did not improve the baseline. It did not isolate replacement of only the 190 distances while retaining all 630 directions.

The main sequence compares the 820 baseline, all 111 additions (931 total), then isolated angle and compatibility-score additions. Grouped permutation analysis shuffles whole related columns jointly on held-out data, five repeats with at most 100 images per letter per fold. Ablations remove a block and refit. These analyses share the fixed full-input model and should not be interpreted as interchangeable measures.

Block weighting tries 0.25, 0.5, 1 and 2. Targeted index MCP/PIP/DIP weighting tries 0.5, 1 and 2. Both use five outer and three inner group folds, with inner training and validation capped at 150 images per letter for cost. The final outer fits use all outer-training images. This is bounded exploratory tuning, not exhaustive hyperparameter optimization.

H.2. Isolated SVM and LightGBM results

Table 7. Detailed original isolated comparison; these results concern matched accepted images.

Representation	Inputs	SVM acc. (%)	SVM macro-F1	LGBM acc. (%)	LGBM macro-F1
Shin	820	97.637	0.97386	97.097	0.96784
Shin + angles	835	97.748	0.97524	97.017	0.96692
Shin + Excel	846	97.648	0.97400	97.119	0.96815
Shin + angles + Excel	861	97.744	0.97515	97.129	0.96824

The full 931-input SVM scored 97.62826% and LightGBM 97.16353%. Removing categorical one-hot columns produced 97.74444% for SVM, matching the isolated angle-plus-workbook model. In the grouped permutation analysis, shuffling

angles reduced SVM macro-F1 by 0.570 percentage points, compared with 0.049 for categories and 0.008 for workbook compatibility scores. For LightGBM these drops were 0.031 and 0.200 for angles and compatibility scores. These overlapping reliance measures are not additive or causal.

The source article reports 87.60% ASL Alphabet combined-feature SVM accuracy, no matching F1 metric, and a held-out evaluation with tuning rather than this filtered grouped protocol. It also states 780,000 images inconsistently with 26 times 3,000; the actual inventory is 78,000. The local paper-profile SVM ($C = 1000$, no balancing) reached 97.66145%; changing this setting alone does not reconcile the protocols. Detection, data selection and split differences have not been isolated experimentally.

I. Tilt Augmentation and Recorded Development Study

The original training input contains 820 Shin values plus 15 angles. For each of the 210 unit direction vectors d_{ij}, the augmentation uses

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}, \qquad d'_{ij} = R_x(\alpha) d_{ij}, \qquad \alpha \in \{-15^\circ, 0, 15^\circ\}.$$

Only input columns 190–819 change (zero-based indexing). The first 190 normalized distances and final 15 intrinsic measurements stay fixed. This rotation is performed in normalized landmark-feature coordinates and is an approximation, not a full physical wrist or perspective model. It does not simulate occlusions or detector errors.

The all-letter run uses 180,753 fitting rows from 60,251 originals, with weight $1/3$ per row. Class balancing and gamma=scale are applied through the fixed SVM pipeline. The C-only run adds two copies of each of the 1,912 C originals, producing 64,075 fitting rows; 58,339 other images remain single examples. Each C source retains total sample weight one, and other rows have weight one. For selective augmentation, preprocessing, balanced class weights and the resolved gamma value are fitted on original fitting rows, avoiding unintended class-frequency or scaling changes. The C-only full-data resolved gamma and class weights match the baseline exactly. The earlier all-letter run computed gamma over its augmented matrix, so the three-way comparison is not a strict single-factor ablation of that numeric parameter.

Both variants were tested on the same original five grouped folds. Augmentation occurs only after splitting, inside the fitting partition. Each variant yields 60,251 original validation predictions. The all-letter variant recovers 57 errors and loses 84 correct predictions: 58,867 correct versus 58,894 for the angle baseline. Its accuracy is 97.70294269% and macro-F1 0.974685870. The C-only variant has the same 58,894 correct images as baseline, with zero recoveries or losses and three wrong-to-wrong label changes; macro-F1 changes from 0.975243912 to 0.975244649. This difference is practically negligible.

The guided development session contains 27 planned captures: C/D/O, three qualitative wrist positions and three repetitions. The intended label and step are fixed; predictions are hidden during collection. All 27 have accepted detections. The session predates candidate fitting and never enters training. The reference Shin-only full-data fit scores 17/27 and C/D/O macro-F1 0.609524; the joint-angle fit scores 17/27 and 0.610035. All-letter tilt scores 18/27 and 0.642439; C-only tilt returns the same predictions as the joint-angle baseline. Joint-angle-model counts by intended letter are C=3/9, D=8/9, O=6/9; all-letter tilt changes only the number correct for O to 7/9. One wrong C prediction also changes from D to O, lowering O precision while improving its recall.

These are development outcomes, not independent test results. The reserved evaluation directory contained zero completed captures when these experiments were run. The previously inspected second-person O example is an exploratory illustration and is not added to the primary 27. No angle/axis search was conducted within either augmentation run, but the research sequence as a whole is exploratory.

The main input/model/image hashes and saved predictions were audited. In the 28 records comprising the development round and extra O example, all image hashes matched and all 835 feature vectors and predictions were reconstructed. The additional reference model is fitted on the same 60,251 original images using only 820 inputs; its live-bundle schema is distinguished from the 835-input models. Snapshot exports retain the selected model's hash, input variant and matching feature width.

Reproduction commands, evidence paths and the live model-label mapping are documented in `REPRODUCE.md`.

J. Later Webcam and Representation Diagnostics

These experiments followed the primary feature comparison. They are retained for transparency and hypothesis generation; they do not expand the main report's research question into a claim that webcam or viewpoint problems have been solved. Stored development errors were reused to motivate successive changes, so their outcomes are not independent final-test evidence.

J.1. Larger G/H Rotations and Projected Angles

A selective G/H augmentation rotates only the 630 image-axis direction values around the camera-y axis by $-90°$, $90°$ and $180°$, retaining each original. There are 4,801 G/H originals and 74,654 fitting rows overall. Each selected source contributes four rows of weight 0.25; other classes occur once with weight one. Preprocessing, balanced class weights and resolved gamma are determined on the original fitting partition. This is a numerical landmark-feature rotation, not a realistic palm/back-of-hand rendering or occlusion simulation.

A separate unaugmented candidate adds 14 internal angles measured from projected image landmarks, after converting normalized x, y into pixels with the detector image width and height. It has 849 inputs: the original 820, 15 estimated 3D measurements and 14 projected angles. The same function is used for training and live inference. Median imputation and standard scaling of all 29 angle inputs are fitted only on each fitting partition. Camera captures are not used for fitting either candidate.

Table 8. Supporting 26-class experiments using the same frozen five folds. These extend the research sequence beyond the main feature comparison.

Model	Correct / 60,251	Accuracy (%)	Macro-F1
Shin + 3D measurements	58,894	97.74776	.97524391
Selective G/H rotation	58,888	97.73780	.97516874
Shin + 3D + projected angles	58,861	97.69298	.97465145

Selective G/H rotation fixes nine formerly wrong predictions and loses 15 correct ones. The projected-angle model fixes 70 and loses 103. Its five-fold accuracy declines in four folds and improves in one. Neither is an overall improvement on the 835-input reference. On the 27 earlier C/D/O development captures, the projected-angle model returns 16 correct versus 17 for the angle baseline.

The selected GH-handrug development round contains 11 errors from the G/H-rotation model: G→H, C→D, F→B, K→H, N→M, O→D, P→H, S→E, T→L, X→H and Y→A. Replaying the original 835-input model on these same landmarks recovers K and X; adding projected angles also recovers T. The newest G remains misclassified. This is a selected-error replay, not a representative live-accuracy comparison.

J.2. G/H-Only Diagnostic Classifiers

The primary 26-class angle SVM recognizes 2,416/2,430 accepted G images correctly (99.424% recall). Its 14 true-G errors are nine J, four H and one F. For H, 2,328/2,371 are correct (98.186%); ten are G and 29 J, with four other errors. Thus strong accepted-image dataset recognition coexists with the recorded webcam failures.

Three fixed binary SVMs were fitted using only 3D angles, only projected angles or both. The same frozen folds were restricted to their 4,801 G/H images. Every variant used training-only median imputation and standard scaling, with RBF $C = 10$, balanced classes and gamma=scale. They are easier two-label tasks with different inputs, not replacements for the 26-class recognizer.

All three fuller binary variants still classify the newest selected G as H. Its middle-PIP estimate is $95.3°$, near the training-G median $86.7°$ and below the H median $166.2°$. Other measurements disagree with typical G: the index PIP is $134.9°$ against a G median $172.6°$, while middle DIP is $161.0°$ against medians $113.2°$ for G and $170.4°$ for H. Its signed thumb deviation is $-1.5°$ against medians around 36–37° for both. The classifier sees a combination, not just a count of straight fingers.

The two-input middle-PIP ablation recovers the two selected G examples and one H example, but has lower grouped

Table 9. Binary G/H diagnostics. The fourth row is an explicitly post-hoc, targeted ablation after inspecting the first three.

Inputs	Number	Accuracy (%)	Macro-F1
3D measurements	15	96.563	.96563
Projected 2D angles	14	95.793	.95791
Both representations	29	96.584	.96584
Middle PIP only, 3D + 2D	2	94.085	.94084

dataset performance. It was proposed after inspecting these known failures and is exploratory. A later selected H with visible palm is called G by all six 26-class variants, although its estimated index/middle PIP angles are $165.5°/159.8°$. The binary diagnostics call it H. This counterexample shows that palm versus back-of-hand viewpoint alone cannot explain every recorded error.

A four-view protocol for three participants (G/H, palm, oblique palm, oblique back-of-hand, back-of-hand; five repetitions) has been implemented, but no completed cohort evaluation is claimed here. The qualitative views are not calibrated physical angles. New balanced and signer-separated observations would be needed for a prospective viewpoint comparison.

Evidence and reproduction commands are in `results/rotation_GH_01/`, `results/angles_2d_01/`, `results/gh_classifier_01/`, `results/gh_middle_01/` and `REPRODUCE.md`. Saved image/model hashes, raw measurements and predictions support computational replay, not anatomical or linguistic ground truth.